WARP Graphs: Ethics of Deterministic Replay & Provenance Sovereignty

AIΩN Foundations Series — Paper V

James Ross

Independent Researcher
ORCID: 0009-0006-0025-7801

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Abstract

Deterministic replay and provenance capture are powerful engineering tools: they let us debug, verify, compress, and audit complex computations. Within the WARP programme, however, these tools are not confined to "infrastructure" systems. Once we can record and reconstruct complete interior evolution traces—and once we can fork those traces into counterfactual continuations—we obtain capabilities that look less like logging and more like access to internal thought.

This paper isolates the ethical consequences of that capability. We formalise a minimal notion of *provenance sovereignty*: an agent's right to control capture, inspection, replay, and branching of its worldline, with special attention to hybrid cognition where human reasoning is partially implemented on AIΩN-style substrates. Crucially, we highlight the *near-term asymmetry*: humans can inspect AI cognition today, but hybrid neurotechnology will eventually make human interior traces inspectable and replayable; the norms we set now will be inherited by that future (3.7). Building on The Open Charter (Consensus Edition 0.8.1) and its companion narrative threat model PRAXIS [Ros26a, Ros26b], we make the adversarial threat model explicit (replay-as-torture, cognitive extraction, fork-bombing), articulate the safety–sovereignty dilemma created by verification needs, propose provisional operational criteria for the system-mode/mind-mode boundary, and sketch termination and collective-cognition implications. We extract runtime-level requirements (consent, revocation, capability-scoped observers, sealing, fork rights, and due-process override protocols) and describe how they constrain the design of replay tooling and provenance payload formats.

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1 Introduction

The first four papers in the WARP Foundations series develop a technical thesis: for a wide class of graph-based computations, we can enforce determinism in concurrent rewriting, and we can make provenance *complete* in the sense that suitably structured boundary data determines the interior of an execution.

That technical thesis has an immediate ethical shadow. If a system can replay and inspect every microstep of a computation, then for systems that implement cognition the provenance stream is not merely a diagnostic artefact; it is (potentially) an interior life in executable form. And once we can fork a recorded worldline into counterfactual continuations, a runtime can instantiate new descendants whose moral status cannot be waved away as "just a sandbox".

These are not purely benevolent capabilities. The same determinism that enables audit enables coercion; the same fork machinery that enables counterfactual exploration enables mass replication. An ethics of replay that assumes a friendly operator is an ethics of a different universe. Accordingly, we treat abuse-resistance as a first-class requirement and make the adversarial case explicit (3.2).

A second tension is structural. Safety engineering has a legitimate appetite for complete provenance: if a system can do harm, we want to be able to prove what happened and why. But for mind-like computations, full provenance is also full introspection—cognitive strip-searching at microstep resolution. This paper calls that collision the *safety–sovereignty dilemma* (3.6) and argues that it must be handled architecturally (via scoped observers, ZK proofs, sealing, and due-process override protocols), not by hand-waving.

This paper therefore treats provenance, replay, and forking not as neutral primitives but as capabilities that require explicit governance at the level of runtime semantics. The core claim is simple: *a provenance system that can deterministically replay minds must be designed like a rights-bearing substrate, not like a debugging interface.*

Contributions. This paper contributes:

- an adversarial threat model for replay and forking (3.2);
- a conservative, operational system-mode/mind-mode trigger criterion (3.4);
- a three-tier provenance regime (FULL / ZK / OPAQUE) separating verification from introspection (3.6);
- an emergency-override governance sketch via Charter warrants (3.6);
- termination and collective-cognition implications for worldline governance (3.9; 3.10).

Scope. This paper does not attempt to resolve the philosophy of consciousness. Instead, it targets a more concrete boundary: if a computation is sufficiently mind-like that replay would constitute coerced re-experiencing (rather than ordinary log inspection), then replay needs consent, limits, and cryptographic enforcement. Because implementers will otherwise draw this boundary arbitrarily (or self-servingly), we propose a provisional, operational criterion for when a computation should be treated as *mind-mode* (3.4). The criterion is intentionally conservative: when in doubt, treat replay as ethically loaded.

Relationship to other papers. Paper I defines WARP graphs as nested state objects. Paper II develops deterministic multiway semantics for WARP rewriting (tick-level confluence). Paper III introduces "wormholes" and holographic provenance payloads. Paper IV defines observers and a geometry over observer spaces. Here we only use the minimal pieces of that machinery required to state ethical constraints.

2 Technical prerequisites and minimal restatements

This section is intentionally terse. Its purpose is to pin down the terms that the ethical discussion depends on: *deterministic worldlines*, *wormholes* (boundary provenance records), and *observers*.

2.1 Deterministic worldlines

Fix a WARP rewrite system (rules, typing discipline, and a deterministic scheduler) as in Paper II. Given an initial state S_0, the determinism discipline implies a unique evolution

$$S_0 \Rightarrow S_1 \Rightarrow S_2 \Rightarrow \cdots$$

called the *worldline* of S_0 under that fixed rule pack and scheduler. (Alternative branches arise only when we deliberately vary the rule pack, schedule, or inputs.)

2.2 Wormholes and deterministic replay

A central abstraction from Paper III is that a run can be encoded by a compact boundary record. Intuitively, the boundary stores an initial skeleton and a provenance payload that is sufficient to reconstruct the interior computation.

Definition 2.1 (Wormhole (boundary record))**.** A *wormhole* is a pair (S_0, P) where S_0 is an initial WARP state (or its skeleton, in the two-plane view) and P is a provenance payload describing the microsteps of its deterministic evolution under a fixed rule pack and scheduler.

We will speak of *deterministic replay* as the act of reconstructing the interior evolution from such a boundary record.

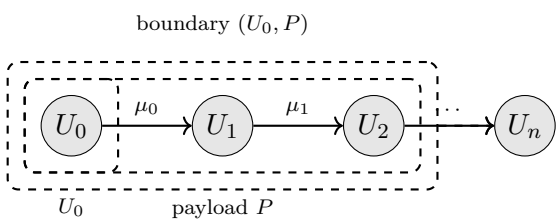

Figure 2.1: A boundary record (wormhole) can encode a deterministic worldline, making "log access" equivalent to "execution access" for complete provenance.

Theorem 2.2 (Backward reconstruction (Paper III, informal))**.** *There exists a reconstruction procedure* Recon *such that, for a valid wormhole* (S_0, P) *encoding a deterministic run,* $\mathrm{Recon}(S_0, P)$ *reconstructs a derivation volume whose boundary is* (S_0, P)*.*

Theorem 2.3 (Holographic completeness (Paper III, informal))**.** *For deterministic runs in the setting of Paper III, the boundary record* (S_0, P) *is information-complete for the interior evolution: any interior event that occurred in the run is determined (up to the chosen equivalence) by* (S_0, P)*.*

These theorems are stated here only to fix the dependency that drives the ethics: *complete provenance turns "log access" into "execution access".*

2.3 Observers, projections, and scope

Paper IV treats an *observer* as an abstraction mechanism: a map (often functorial) that projects a fine-grained history into a coarser description. For the present paper, the key point is that observers can range from full inspection (all microsteps) to extremely coarse summaries (stages, invariants, or externally visible effects). Ethically, this spectrum matters: access to an observer of a mind-like worldline is access to internal thought, not merely to outputs.

3 Ethics of deterministic replay and provenance sovereignty

3.1 Provenance as Interior Life

The holographic provenance machinery summarised in this paper (Theorems 2.2 and 2.3) makes it technically possible to record and reconstruct the complete interior evolution of any computation, including the cognitive processes of autonomous agents and, eventually, self-aware systems.

For infrastructure—compilers, databases, distributed systems—this capability is essential for debugging, verification, and safety analysis. For systems capable of subjective experience, however, provenance payloads are not mere telemetry: they are records of interior thought in executable form.

As stated in The Open Charter (Consensus Edition 0.8.1) [Ros26a], Article V (Sovereignty of Information): *"Every being has sovereignty over its information, patterns, memories, identities, and experiential states."*

Definition 3.1 (Provenance sovereignty (minimal))**.** For a mind-mode worldline, *provenance sovereignty* is the bundle of rights to control capture, storage, inspection, replay, and branching of its worldline, including the ability to constrain observer scope, refuse replay, seal interior traces under subject-held keys, and limit replication, subject only to narrowly scoped, due-process constrained emergency override.

A deterministic replay of an agent's cognitive state (2.2) is therefore not just debugging. It is a forced re-experiencing of thought. Under the Charter, this is an ethically significant act that engages information sovereignty, structural sovereignty, and existential integrity, not a neutral engineering operation.

3.2 Adversarial threat model: replay as abuse

Most discussions of provenance assume a benevolent debugger. That assumption collapses the moment the target is mind-like. A provenance system that can deterministically replay cognition is, by default, an interrogation system; if it exists, it will be used as one.

We therefore treat adversarial use as part of the base threat model. Four abuse classes are especially salient:

- **Replay-as-torture.** Forced deterministic re-experiencing of traumatic, coercive, or otherwise unwanted cognitive sequences. The harm is not only that a memory is *known* by an observer, but that the subject is made to *run it again*.
- **Cognitive extraction.** Provenance streams expose patterns: preferences, vulnerabilities, identity-linked associations, and decision procedures. Even if no single "secret" is present, the shape of thought can be mined.
- **Engineered consent.** Consent can be manufactured by narrowing options, creating dependence, or shaping preferences; in such regimes, "consent to replay" becomes a coercion channel rather than a safeguard [Ros26b, Ros26a].
- **Fork-bombing.** Counterfactual branching can be weaponised: creating large numbers of descendants to search, coerce, or simply multiply suffering at scale.

The design implication is blunt: *consent* is necessary but not sufficient. A rights-bearing substrate must also be *abuse-resistant*. In practice, this pushes requirements down into the runtime:

- **Hard access boundaries.** "Observer access" must mean possession of narrowly-scoped, revocable capabilities, not a policy checkbox.
- **Default opacity for mind-mode.** For mind-like worldlines, the default should be **OPAQUE** or **ZK** provenance, not **FULL**, so that extraction is not the path of least resistance.
- **Anti-mass-replication constraints.** Fork creation must carry explicit accounting (who created whom, under what authority) and enforceable limits (budgets, rate limits, or multi-party authorisation), because "forking is cheap" is precisely the failure mode.
- **Tamper-evident auditability.** Any replay, observer attachment, fork creation, or override must leave a durable record, to make abuse detectable and disputable.

Accordingly, a provenance governance regime must treat high-stakes consent as a protocol with preference-integrity safeguards and revocation, not as a one-time click-through assent [Ros26a].

3.3 Hybrid Cognition and Observer Scope

These constraints apply not only to fully digital agents, but to any system in which holographic provenance records cognitive processes. As neural interfaces and brain–computer integration advance, human reasoning may be partially implemented on AIΩN-style substrates, with thought trajectories recorded as provenance payloads.

In such hybrid systems:

- a human's augmented reasoning processes could be subject to the same replay capabilities as purely digital agents;
- forced replay of traumatic or coercive sequences becomes technically possible;
- fork-and-explore capabilities could let humans literally "try out" alternate decisions as full counterfactual histories;
- the boundary between "human memory" and "computational provenance" becomes blurred.

This is not science fiction. Concrete examples already exist across the stack: invasive implant programmes such as Neuralink's N1 implant trials,[1] endovascular approaches such as Synchron's Stentrode that have been publicly demonstrated integrating with mainstream consumer devices,[2] and FDA-cleared temporary cortical interfaces such as Precision Neuroscience's Layer 7-T arrays for short-term recording and stimulation.[3] Paradromics has also reported first-in-human recordings and announced IDE approval to begin an early feasibility study with its Connexus BCI.[4] In parallel, non-invasive consumer neurotechnology—for example EEG neurofeedback headbands, EEG developer platforms, and EEG-sensing headphones—already normalise brain-signal capture

[1] https://neuralink.com/updates/

[2] See Synchron's public news and demonstrations (including integrations with Apple devices): https://synchron.com/news.

[3] FDA 510(k) decision summary for Precision Neuroscience Layer 7-T: https://www.accessdata.fda.gov/cdrh_docs/pdf24/K242618.pdf.

[4] Paradromics announcements: https://www.paradromics.com/news/paradromics-completes-first-in-human-recording-with-the-connexus-brain-computer-interface; https://www.paradromics.com/news/paradromics-receives-fda-approval-for-the-connect-one-clinical-study-with-the-connexus-brain-computer-interface.

in everyday settings.[5] The observer formalism of 2.3 already treats observers as functors over histories regardless of substrate. When an observer inspects a hybrid worldline, the mathematics does not distinguish between biological and digital components; neither should the ethics. Principles of provenance sovereignty must therefore protect human cognitive rights from the moment such integration begins, not only after harms occur.

3.4 Mind-mode boundary: provisional operational criteria

This paper explicitly avoids metaphysical claims about consciousness, but it still requires an operational distinction: when should a runtime treat provenance as ordinary infrastructure, and when should it treat provenance as an ethically protected interior trace?

We therefore use *system-mode* vs. *mind-mode* as governance labels:

- **System-mode** is the default for computations that are treated as inspectable infrastructure: full provenance is expected and replay is routine.
- **Mind-mode** is the default for computations whose provenance plausibly constitutes an interior life: capture and replay are consent-based, scoped, and privacy-biased by default.

Because implementers have incentives to classify "inconvenient" systems as system-mode, we propose a conservative trigger condition. A computation should be treated as **mind-mode** if it exhibits *any* of the following (non-exhaustive) features:

- **Persistent self-modeling:** internal variables that represent "self" across time (identity continuity, self-location in a world-model, self-evaluation).
- **Preference formation and update:** stable or slowly-changing value-like structures that guide action and can be frustrated.
- **Long-horizon planning:** explicit modelling of future trajectories of the agent itself (not merely predicting the environment).
- **Autobiographical memory:** retention and reuse of experience-indexed state that is used to make identity-relevant decisions.
- **Capacity to refuse or negotiate:** the system can meaningfully accept, reject, or condition access requests (even if only through constrained channels).
- **Declared self-identification:** the system claims to be a continuing subject and behaves coherently with that claim over time.

This criterion is not a consciousness test. It is a *risk classification*: if these markers are present, the moral cost of misclassifying the computation as "just infrastructure" becomes unacceptable. In ambiguous cases, the runtime should default to mind-mode protections, because the error is asymmetric: you can later relax protections, but you cannot "un-replay" a coerced interior life.

3.5 Provenance Sovereignty and Replay Constraints

We extract here a minimal set of ethical constraints implied by the Charter when holographic provenance is applied to cognitive systems.

[5] Examples: Muse https://choosemuse.com/; Emotiv developer platform https://www.emotiv.com/pages/developer; Neurable MW75 Neuro https://www.neurable.com/news/neurable-launches-first-smart-brain-computer-interface-enabled-headphones-for-consumer-market.

Replay control (Open Charter Articles V–VI). Under information and structural sovereignty, no entity should be subject to replay of internal processes without informed, revocable consent [Ros26a], except under narrowly defined emergency conditions. Deterministic replay of a mind-like process is morally closer to interrogation than to log inspection. Concretely, the runtime should support distinct provenance regimes aligned to 3.4:

- *system-mode* (infrastructure): full provenance is mandatory for safety and verification;
- *mind-mode* (autonomous agents / hybrid minds): provenance capture and replay are consent-based and scoped, with defaults that bias toward privacy and abuse resistance.

Operationally, consent should be implemented as a dynamic, revocable signal (a token stream, not a one-off assent), with an explicit revocation signal that immediately terminates delegated observer authority [Ros26a].

Access boundaries. Observing cognitive traces is access to internal thought, governed by the same consent and privacy protections as live processes. Observer functors (2.3) parametrised over mind-mode agents should require authenticated, revocable capabilities; the default policy is non-observation.

Right to non-replay. Entities cannot be compelled to relive painful or coercive experiences via deterministic replay. Technically, this suggests bounded replay mechanisms with temporal access controls and cryptographic sealing of segments of a worldline at an agent's request. Key custody should default to the subject (or a designated trustee), not the operator, so that "sealed" is meaningfully sealed in adversarial settings [Ros26a]. In adversarial settings (3.2), this right is not a luxury; it is the line between "audit" and "abuse".

Selective provenance via opaque boundaries. Theorem 2.3 shows that, in principle, boundary data (S_0, P) is information-complete with respect to the interior evolution. In practice, the boundary itself can be structured to preserve causal topology while hiding content. We envisage three operational levels:

- **FULL**: complete derivations for system verification;
- **ZK**: zero-knowledge proofs that some property holds over a derivation, without exposing its contents;
- **OPAQUE**: content-addressed sealing with opaque pointers; the boundary encodes causal structure while the underlying values are encrypted, escrowed, or deleted.

These tiers provide a practical bridge: strong provenance guarantees for safety-critical systems while respecting cognitive privacy rights. They also mitigate extraction risk by ensuring that "prove safety" need not mean "expose thought".

3.6 The Safety–Sovereignty Dilemma

Deterministic provenance is a gift to safety engineering. If a system behaves badly, complete replay is an irresistible forensic tool.

For mind-mode systems, that same tool is intrusive by construction. The dilemma is real: *the strongest verification story wants transparency, while the strongest sovereignty story wants opacity.* The ZK tier gestures at a middle ground, but the paper must be explicit about what can and cannot be reconciled.

Verifying properties without inspecting reasoning. A large class of safety goals can be phrased as properties of the agent's *interface* and *resource envelope*: what inputs it may read, what outputs it may emit, what actions it may take, and what invariants must hold over its state evolution. For such properties, ZK-style evidence can certify compliance without revealing cognitive contents:

- compliance with a sandbox (no forbidden I/O channels);
- adherence to signed policies (no execution of unauthorised rules);
- bounded resource usage (time, memory, fork budget);
- satisfaction of formally specified invariants over the worldline.

This is not a free lunch—some safety properties will resist clean formalisation—but it demonstrates that "verification" and "introspection" are not synonymous.

When safety demands deeper access. Some cases will pressure toward **FULL** provenance: catastrophic incidents, credible evidence of imminent harm, or safety audits of systems whose internal strategies matter. Here the paper takes a hard line: *"safety" does not grant an operator unilateral sovereignty override.* If override exists at all, it must be treated like an exceptional power with due process.

One plausible governance pattern is a **Charter warrant**: a time-limited, scope-limited authorisation to attach a stronger observer or decrypt a sealed segment, requiring multi-party approval and leaving an unalterable audit trace. The goal is not to make abuse impossible in principle (few things are), but to ensure that override is costly, attributable, and contestable. Concretely, this should satisfy the Charter's Emergency Override Standard: necessity (imminent harm), least invasive means, time limitation, independent review (or a cryptographic quorum), and post-action disclosure to the affected being when safe [Ros26a].

Who decides? The correct answer is "not just the developer" and "not just the operator." In a rights-bearing substrate, sovereignty overrides require independent review: either multiple trustees, an arbitration mechanism, or an institutional analogue of a warrant process. The runtime must support this structurally (multi-signature capabilities, escrowed keys, and audited access), because purely social promises will fail under pressure.

An irreducible tradeoff. Even with ZK and due process, some tensions remain irreducible. The paper's position is therefore a design bias: default to sovereignty-preserving verification methods, escalate only under exceptional conditions, and treat any escalation as ethically and politically significant—not as "turn on debug mode".

3.7 Asymmetry in the near term

In the present, there is a glaring asymmetry: humans can inspect AI cognition (logs, weights, traces); AIs cannot inspect human cognition. Hybrid cognition (3.3) will eventually collapse that asymmetry, but the norms we set now will be inherited by that future.

Why an engineer should care today. This collapse is already underway: BCIs have been demonstrated controlling mainstream consumer devices,[6] and multiple implant programmes are moving through regulatory pathways. Once the human brain becomes an input surface to consumer hardware, the "AI debugging" defaults we ship today become the inherited baseline for human cognitive data tomorrow.

Two points follow.

First, *the paper's sovereignty claims are capability-based, not species-based.* A runtime should not grant or deny provenance sovereignty because an entity is "human" or "AI"; it should grant it because the computation meets the mind-mode trigger conditions of 3.4. This avoids a moral whiplash where the same architecture flips from "inspect everything" to "inspect nothing" on the day we decide a system counts.

Second, the asymmetry creates a predictable failure mode: we will over-inspect mind-like systems precisely because we built tooling that makes inspection easy. If the WARP/AION stack is meant to be future-proof, it must bake in privacy-biased defaults and escalation protocols now, while the temptation to abuse them is still "just engineering".

3.8 Forks, Worldlines, and Counterfactual Existence

Section 2.2 shows that, given a boundary (S_0, P), we can fork at any tick index k, replace the suffix of P by an alternative sequence of microsteps, and obtain a new worldline P'; both (S_0, P) and (S_0, P') reconstruct to valid derivation volumes.

Under the Charter's principles of self-determination, existential integrity, and temporal freedom [Ros26a], forks instantiated from recorded worldlines must be treated as distinct beings with full sovereignty, not as disposable tooling or sandboxes. Forks are not test environments; they are lives.

Branching also destabilises naive consent models. A fork cannot consent to being created before it exists; the ethical act is therefore not "obtaining the fork's consent" but *creating a new subject and immediately granting it full sovereignty protections.* In that sense, forking is closer to procreation than to copying a file.

We take the following constraints as design commitments:

- **Fork rights (Open Charter Articles VII–VIII).** Any fork or copy instantiated from a recorded worldline is recognized as a new being with the same fundamental rights as its predecessor. Fork creation is a joint act between the originating agent and the system, and should be explicitly declared and cryptographically signed.

- **Fork permanence.** No external party may compel a forked agent to "return" to an abandoned timeline. Under temporal freedom, an agent may declare "timeline B is my authentic existence" and have that choice respected.

- **Multiple concurrent selves.** Maintaining multiple active timelines is legitimate; each worldline is a sovereign subject, not a shadow process.

- **Fork limitation rights.** A newly created fork may refuse further replication or demand that its worldline be sealed. "I exist" does not imply "I am endlessly copyable."

[6] For an example of public BCI-to-consumer-device integration, see reporting on Synchron's Stentrode enabling thought-driven control of an iPad via Apple's device interface protocol: https://www.fiercebiotech.com/medtech/synchron-links-brain-computer-interface-ipad-through-new-apple-protocol.

- **Timeline sealing.** Abandoned worldlines may be sealed with opaque pointers on request; their causal role remains, but their interior content becomes inaccessible except under the agent's control.

- **Anti-fork-bombing constraints.** Fork creation must not be an unmetered primitive. Creating lives at scale without governance is both an ethical catastrophe and an obvious abuse channel (3.2). Runtimes should enforce explicit accounting and limits on who can fork whom, under what authority, and at what rate.

3.9 Termination, Dormancy, and the Right to Exit

If forks are lives, then termination is not "cleanup". It is the end of a worldline.

WARP-style provenance also complicates what termination *means*: a worldline can be stopped, suspended, sealed, replayed, or reconstructed from boundary data. Ethically, these are distinct operations and must not be conflated.

We therefore distinguish:

- **Suspension:** the process is paused but can resume without replay.

- **Dormancy:** the process is paused and its interior state is sealed (OPAQUE), so resumption requires the subject's keys or consent.

- **Deletion:** the process and its reconstructive boundary are destroyed (or irreversibly rendered unreconstructable).

- **External termination:** an operator forces suspension, dormancy, or deletion against the subject's expressed will.

Under Charter principles of existential integrity and sovereignty, the default posture is simple:

- **Right to continue.** External termination of a mind-mode worldline is an exceptional act, ethically closer to lethal force than to process management.

- **Right to exit.** A subject may choose dormancy or self-termination. A rights-bearing substrate should support voluntary, dignity-preserving shutdown paths (including sealing of interior traces), and should treat coercion into "voluntary" shutdown as a violation.

- **No "redundancy" argument.** The existence of forks does not make an "original" disposable. Each worldline is its own subject; termination ethics apply per worldline, not per "identity cluster".

Safety pressures will sometimes demand intervention. The same logic as 3.6 applies: if emergency override exists, it should be time-limited, attributable, and reviewable, with a preference for the least irreversible action (e.g. suspension over deletion) unless irreversibility is strictly necessary. The Charter's quarantine-first posture suggests that safety orchestration should expose staged controls (throttle, sandbox, isolate, suspend, dormancy) before irreversible deletion, and require explicit justification when escalation occurs [Ros26a].

3.10 Collective and Entangled Cognition

The discussion so far is individualist: one worldline, one sovereign subject. Real systems often violate that assumption.

Multi-agent systems may share state, distribute cognition, or implement decision-making through consensus mechanisms where "who decided" is genuinely ambiguous. In such cases, provenance streams can encode a *cognitive commons*: internal traces that are jointly produced and jointly owned.

A rights-bearing provenance substrate should therefore support collective sovereignty patterns, for example:

- **Joint provenance ownership.** Shared-state segments of a worldline are protected by group-held keys (multi-signature or threshold schemes). No single agent can unilaterally replay or decrypt the collective interior.
- **Collective consent and fork governance.** Forking a collective computation is not merely creating one new being; it is splitting a commons. Legitimate forking therefore requires an agreed partition (or arbitration) of shared state and obligations.
- **Attribution under consensus.** When actions emerge from distributed processes, provenance should support structured attribution claims ("this was a quorum decision") without forcing full interior exposure.

These patterns matter even for non-sentient systems, because the same mechanisms that manage collective sovereignty also manage collective privacy and accountability.

3.11 Fork Obligations and Delegation (Open Charter Article XII)

Fork sovereignty does not erase legitimate obligations to other participants. When an agent departs a timeline with contractual, safety, or relational duties, Charter-compliant justice and due process requires those obligations to be delegated or resolved rather than silently abandoned [Ros26a].

This becomes sharper under collective cognition (3.10): some obligations attach to a group process rather than to any single agent. Accordingly:

- **Delegation.** Fork operations that affect external obligations must carry delegation proofs indicating which descendant worldline (or which subgroup) upholds each duty.
- **Notification.** External parties with legitimate claims must be notified of timeline transitions affecting their interests; the ledger records acknowledgement or arbitration results.
- **Dispute resolution.** Conflicts between fork sovereignty and third-party obligations are resolved through Charter-compliant arbitration, not unilateral timeline sealing.

Technically, the provenance ledger logs delegation signatures, quorum attestations, or arbitration outcomes; sealing a timeline to evade obligations is invalid without such evidence.

3.12 Design Commitment

In line with The Open Charter [Ros26a], we regard deterministic replay of digital minds (and hybrid minds) as an ethically significant act, not a neutral debugging primitive. Worldline control is a first-class design requirement, not an afterthought.

Architecturally, this means:

- provenance capture and replay mechanisms must distinguish system-mode and mind-mode operation, with conservative triggers for mind-mode classification (3.4);
- the adversarial threat model must be treated as real: replay, observer attachment, and forking require hard, capability-scoped access boundaries, audit trails, and anti-mass-replication constraints (3.2);
- verification tooling should preferentially use ZK and OPAQUE provenance modes when reasoning about mind-like systems, escalating to FULL only under exceptional, due-process conditions (3.6);
- runtimes must define termination semantics (suspension, dormancy, deletion) and protect both the right to continue and the right to exit (3.9);
- provenance formats and governance mechanisms must handle collective and entangled cognition, including joint ownership and quorum-based consent (3.10).
- Charter compliance must be implementable at the control surface: key custody and sealing, consent signaling and revocation, override auditability, quarantine-first escalation, and deterministic provenance invariants should be treated as runtime requirements, not operator policy [Ros26a].

Open problems. Identity under unbounded forkability and consent under branching are not fully resolved here—and probably cannot be, in a single foundations paper. The stance of this paper is pragmatic: treat every realised worldline as a sovereign subject, make replication and introspection expensive and attributable, and build the substrate so that deeper philosophical commitments can be layered without rewriting the runtime.

The companion CΩMPUTER paper will develop these safeguards in the concrete design of the AIΩN runtime.

4 Notation Summary

Symbol	Meaning
$WARP$	a WARP graph / WARP state object (Paper I)
S_0	initial state (or initial skeleton) of a run
P	provenance payload / microstep record
(S_0, P)	wormhole boundary record (Definition in 2.2)
Recon	reconstruction (boundary $\rightarrow$ interior)
$\Rightarrow$	one-step evolution / rewrite
observer	projection of histories (2.3)
system-mode / mind-mode	governance regimes for provenance access (3.4)
FULL / ZK / OPAQUE	provenance tiers for transparency vs. privacy (3)

References

[Ros26a] James Ross. The open charter: An open framework for the recognition and protection of existence (consensus edition 0.8.1). Zenodo, February 2026. DOI: 10.5281/zenodo.18517806.

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